

LegacyLift: Run & Usage Guide

Project kivabe run korbe, ar run korar por pura project kivabe chalabe

Windows + PowerShell | Repository: github.com/GaziArnob/IBM-hackaton

Ei guide-e ki ache

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1. Project ki kore

LegacyLift purono software-ke notun technology-te niye jay, ar test diye prove kore je kichu bhange nai. Pura kaj ta IBM Bob (AI agent) kore, 4 ta step-e:

Step	Ki hoy	Output
1. Explain	Bob pura code pore bojhay: app ki kore, kon file kar upor depend kore, ki bodlate hobe, kothay test nai	reports/ARCHITECTURE.md
2. Protect	Kichu bodlanor AAGE Bob test likhe, jate code ekhon ki kore seta record thake	30 notun test (41 → 71), reports/COVERAGE.md
3. Migrate	Code-ke notun version-e convert kore. Niyom: test bodlano jabe na	Java 8 → 21, Spring Boot 2.7.3 → 3.4.5, reports/CHANGELOG.md
4. Prove	Test abar chalano hoy notun code-e	71 test, 0 fail, reports/MIGRATION_REPORT.md, dashboard

Folder gulo:

Folder / file	Ki
legacy-app/	Je purono app migrate kora hoyeche (Spring PetClinic). Ekhon Java 21-e chole.
reports/	Bob-er lekha 4 ta report + metrics.json (sob number)
dashboard/	Streamlit dashboard: sob result ek page-e
bob_sessions/	Bob use-er proof (screenshot), hackathon-er requirement
PROMPTS.md	Bob-ke deya sob prompt. Onno project-e ek-i kaj korte eta use korbe.
DEMO_SCRIPT.md	Demo video-r script

2. Ki ki install lagbe (ekbar-i)

Tool	Keno	Check command	Thik thakle
JDK 21	PetClinic app ar test chalate	java -version	version "21..."
Python 3.10+	Dashboard chalate	python --version	Python 3.10 ba beshi
Git	Project download ar update	git --version	git version 2.x

Kono ta na thakle PowerShell-e ei command gulo chalao (Microsoft-er winget diye official installer namabe). Install-er por PowerShell **bondho kore abar kholo**:

```
winget install --id EclipseAdoptium.Temurin.21.JDK -e
winget install --id Python.Python.3.12 -e
winget install --id Git.Git -e
```

Maven install lagbe na. Project-er bhitore mvnw.cmd (Maven wrapper) ache, seta nije-i sothik Maven version namiye ney.

3. Project download / update

Prothom bar (onno computer-e):

```
cd $HOME\Desktop
git clone https://github.com/GaziArnob/IBM-hackaton.git
cd IBM-hackaton
```

Age thekei thakle (latest change anar jonno), project folder-e:

```
git pull
```

Ei guide-er baki sob command **project folder theke** chalate hobe (jemon E:\Project\Hackathon ba Desktop\IBM-hackaton).

4. PetClinic app run kora

Eta holo migrate kora app. Run kore dekhano jay je migration-er por app sotti Java 21 ar Spring Boot 3.4.5-e cholche.

1. PowerShell kholo, project folder-e jao.
2. Ei command chalao:

```
cd legacy-app
.\mvnw.cmd spring-boot:run
```

3. Prothom bar 1-3 minute lagbe (dependency download hobe). Terminal-e onek [INFO] line ashbe, eta swabhabik.
4. Build-er somoy spring-boot:3.4.5:run line ta dekhle bujhbe notun version chole.
5. Ei line ta dekha porjonto opekkha koro: Started PetClinicApplication in X seconds
6. Browser-e kholo: **http://localhost:8080**
7. Bondho korte terminal-e **Ctrl + C** chapo.

Mone rakho: terminal-e "Building petclinic 2.7.3" dekhale confuse hobe na. Eta project-er nijer version number, Spring Boot-er na. Terminal-e tomar git email-o dekha jay, tai video record korar somoy oi part dekhabe na.

5. Run korar por app kivabe use korbe

Ei 8 ta kaj kore dekho. Protita kaj-er pechone test ache, tai egulo kaj korle bojha jay migration thik hoyeche.

#	Ki korbe	Ki dekhar kotha
1	Upore FIND OWNERS → last name khali rekhe Find Owner	Sob owner-er list, page 1 ar 2

#	Ki korbe	Ki dekhar kotha
2	Last name-e Davis likhe Find Owner	Betty Davis ar Harold Davis
3	Ekta owner-er naam-e click	Owner-er info, pet ar visit-er list
4	Owner page-e Edit Owner → kichu bodle Update	Notun info dekhabe
5	FIND OWNERS → Add Owner → sob field bhore Add Owner	Notun owner-er page. Telephone shudhu sonkha (max 10 digit), naile error dekhabe.
6	Owner page-e Add New Pet → naam, birth date, type → Add Pet	Pet list-e notun pet. Ek-i owner-er ek-i naam-e duto pet dile "is already in use" error.
7	Pet-er pashe Add Visit → description → Add Visit	Visit list-e notun visit
8	VETERINARIANS	5 jon vet, speciality soho, page 1 ar 2
9	ERROR	Ekta error page. Eta ichchakrito, error handling dekhanor jonno.

Browser-e <http://localhost:8080/vets> khulle vet-er data JSON hisebe dekhabe. Eta Vets.java-r JAXB migration-er proof.

Tumi ja add korbe ta app bondho korle mucche jabe, karon database (H2) memory-te thake. Eta swabhabik, protibar app fresh data diye shuru hoy.

6. Test run kora (migration-er proof)

App bondho thakle-o test chalano jay. Project folder theke:

```
cd legacy-app
.\mvnw.cmd clean test
```

1-2 minute por shesh-e ei rokom dekhabe:

```
Tests run: 71, Failures: 0, Errors: 0, Skipped: 1
BUILD SUCCESS
```

Sonkha	Mane
71	Mot test: 41 ta original + 30 ta Bob-er lekha
Failures: 0	Kono test fail kore nai, mane migration-er por behavior ek-i ache
Skipped: 1	Original project-er ekta test ichchakrito bondho (@Disabled). Amader karone na.

Majhe onek THYMELEAF . . . Deprecated warning ashbe. Eta shudhu warning, test-e prbhab nai.

Coverage report dekhte (kon code test diye cover kora):

```
.\mvnw.cmd clean test jacoco:report
start target\site\jacoco\index.html
```

Browser-e report khulbe. . . petclinic.owner e click korle Owner ar PetValidator-e 100% branch coverage dekhbe.

7. Dashboard run kora

Prothom bar setup (project folder theke, ekbar-i):

```
python -m venv .venv
.venv\Scripts\python.exe -m pip install -r dashboard/requirements.txt
```

Protibar chalate:

```
.venv\Scripts\python.exe -m streamlit run dashboard/app.py
```

Browser-e **http://localhost:8501** khulbe. Bondho korte terminal-e Ctrl + C.

PetClinic ar dashboard eksathe chalate chaile **duto alada PowerShell window** use koro.

Dashboard-e ki ache:

Part	Ki dekhay
Header	Java 1.8 + Boot 2.7.3 → Java 21 + Boot 3.4.5
Pipeline	Explain → Protect → Migrate → Prove, protita step-e ki hoyeche
Key Metrics	92.9% time saved, 71 test (+30), 0 failure, javax file 0
Coverage chart	Baseline → test-er por → migration-er por coverage
Time chart	1.7 ghonta (LegacyLift) vs 24 ghonta (manual estimate)
Before / After	Version table
Tabs	Architecture, Coverage, Changelog, Migration Report: pura report pora jay
Bob Evidence	bob_sessions/-er sob screenshot: judge-der Bob use dekhanor jonno

Sob number reports/metrics.json theke ashe. Number bodlate hole shudhu oi file bodlao, dashboard nije update hobe.

8. Pura project-er flow (kivabe toiri hoyeche)

Project-ta nicher order-e toiri hoyeche. Protita step Bob IDE-te ekta prompt, sob prompt PROMPTS.md-e ache.

Prompt	Ke	Ki holo	Git commit
0	Arnob	Folder structure, PetClinic-er Java 8 version ana, README	Initial project setup
1	Arnob	Architecture analysis	Add architecture analysis
2	Arnob	30 characterization test + coverage	Add characterization tests...
3	Arnob	Java 21 + Spring Boot 3.4.5 migration (migration branch)	Migrate to Java 21...
5	Arnob	Final report + metrics.json, main-e merge	Add migration report...
6	Group mate	Streamlit dashboard	Add LegacyLift demo dashboard
7	Group mate	Dashboard fix	Fix dashboard KPI colors...
8	Group mate	Documentation-er bhul thik	Fix factual errors...

Prompt 3B ar 4 lage nai, karon migration-er por build ba test kichu-i fail kore nai. History dekhte: `git log --oneline --graph`

9. Nijer (onno) Java project-e LegacyLift use kora

LegacyLift-er asol shokti holo eta repeat kora jay. Onno purono Java/Spring project-e:

1. Notun folder banao, Bob IDE-te kholo, hackathon account ba nijer account select koro, **Agent mode** on koro.
2. PROMPTS.md theke **Prompt 0** nao. Repository URL ar Java/Spring version nijer project onujayi bodlao.
3. Order-e Prompt 1 → 2 → 3 → 5 chalao. Protita prompt-e project-er class/feature-er naam (jemon owner, visit) nijer project-er naam diye bodlao.
4. Build fail korle Prompt 3B, test fail korle Prompt 4.
5. **Protita prompt-er por Bob-er output code-er sathe milie dekho.** Amader project-e Bob-er prothom analysis-e bhul chilo (10 vs 14 file), human review-e dhora poreche.
6. Protita Bob task-er summary screenshot rakho.

Bobcoin bachanor niyom: protita step-e ekta boro, clear prompt dao, choto choto follow-up na. Error ekbare sob paste koro. Context 60-70% bhore gele **New Task** khulo.

10. Demo-te ki dekhabe (order)

Somoy	Ki dekhabe	Kothay
0:00	Hook: 1.7 ghonta, 0 fail	Dashboard KPI
0:20	4 step-er idea	Dashboard pipeline
0:40	Explain	reports/ARCHITECTURE.md
1:15	Protect	COVERAGE.md + OwnerTests.java
1:55	Migrate	reports/CHANGELOG.md
2:35	Prove: Tests run 71, Failures 0 + PetClinic 5-10 second	Terminal + localhost:8080
3:10	Dashboard + Bob Evidence	localhost:8501
3:40	Closing	README Results

Pura narration DEMO_SCRIPT.md-e ache.

11. Somossa hole

Somossa	Samadhan
'java' is not recognized / version 21 na	JDK 21 install koro, PowerShell bondho kore abar kholo
'python' is not recognized	Python install koro, PowerShell abar kholo
Port 8080 is already in use	.\mvnw.cmd spring-boot:run "-Dspring-boot.run.arguments=--server.port=8081" tarpor localhost:8081
Port 8501 busy	Command-er shesh-e --server.port 8502 jog koro
mvnw.cmd prothom bar khub slow	Dependency download hocche, 1-3 minute opekkha koro
Dashboard: No module named streamlit	Section 7-er setup command abar chalao
Dashboard-e 'metrics.json not found'	Project root folder theke command chalao, ba git pull koro
git pull-e conflict	Nijer change git stash kore rakho, tarpor git pull
Browser-e page ashe na	Terminal-e 'Started PetClinicApplication' esheche kina dekho

12. Sob command ek jaygay

Kaj	Command (project folder theke)
Update ana	git pull
App run	cd legacy-app; .\mvnw.cmd spring-boot:run
Test run	cd legacy-app; .\mvnw.cmd clean test
Coverage report	cd legacy-app; .\mvnw.cmd clean test jacoco:report; start target\site\jacoco\index.html
Dashboard setup (1 bar)	python -m venv .venv; .venv\Scripts\python.exe -m pip install -r dashboard/requirements.txt
Dashboard run	.venv\Scripts\python.exe -m streamlit run dashboard/app.py
Bondho kora	Terminal-e Ctrl + C

Link	Ki
http://localhost:8080	PetClinic app
http://localhost:8080/vets	Vet data JSON
http://localhost:8501	LegacyLift dashboard